

For the first and second exercises, you can consult
<https://youtu.be/sXuhGPwM7T8>, and <https://youtu.be/eDfQAWcf87Y>.

1. Let a, b, c, d be real numbers and $c \neq 0, d \neq 0$, prove that

- $ac = bc$ implies $a = b$.
- $a0 = 0$
- $-(-a) = a$
- $(c^{-1})^{-1} = c$
- $(-1)a = -a$
- $a(-b) = -(ab) = (-a)b$
- $(-a) + (-b) = -(a + b)$
- $(-a)(-b) = ab$
- $(a/c)(b/d) = (ab)/(cd)$
- $(a/c) + (b/d) = (ad + bc)/cd$

2. Show that, if a, b and c are real numbers, then

- $a < b$ and $c < 0$ implies $ac > bc$
- $0 < 1$ and $-1 < 0$
- $a > 0$ implies $1/a > 0$
- $0 < a < b$ implies $0 < 1/b < 1/a$

3. Solve in $\mathbb{R}$ the inequalities, while determine the domain of validity:

$$x - 1 \leq \sqrt{x + 2}; \quad (2x - 7) \ln(x + 1) > 0; \quad \ln\left(\frac{x+1}{3x-5}\right) \leq 0, \quad x + \frac{1}{x} \geq 2.$$

4. Prove that for any $x > 0$, we have

$$x - \frac{x^3}{6} \leq \ln(x + 1) \leq x.$$

5. Let a, b be real positive numbers. Show that

$$a^2 + b^2 + c^2 \geq ab + bc + ca.$$

Then

$$(a + b + c)^2 \geq 3(ab + bc + ca).$$

6. Let a, b be real nonnegative numbers. Show that

$$8(a^4 + b^4) \geq (a + b)^4.$$

7. (a) Let $k = 1, \dots, n$, and let $a_k > -1$ be real numbers that are all non-negative or all non-positive. Show that

$$(1 + a_1)(1 + a_2) \dots (1 + a_n) \geq 1 + a_1 + \dots + a_n.$$

(b) Deduce that for any $a > -1$

$$(1+a)^n \geq 1+na. \quad (\text{Bernoulli's inequality})$$

8. Let M_a, M_g, M_h denote the arithmetic, geometric, harmonic means of n positive real numbers $x_1, x_2, \dots, x_n$, respectively, where

$$M_a := \frac{\sum_{i=1}^n x_i}{n}, \quad M_g := \left(\prod_{i=1}^n x_i \right)^{1/n}, \quad M_h := \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}.$$

Show that

$$M_h \leq M_g \leq M_a.$$

9. Let $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ be real numbers. If $a_1 \geq a_2 \geq \dots \geq a_n$ and $b_1 \geq b_2 \geq \dots \geq b_n$ (or $a_1 \leq a_2 \leq \dots \leq a_n$ and $b_1 \leq b_2 \leq \dots \leq b_n$), show that

$$(a_1 + a_2 + \dots + a_n)(b_1 + b_2 + \dots + b_n) \leq n(a_1 b_1 + a_2 b_2 + \dots + a_n b_n). \quad (\text{Chebyshev's inequality})$$

10. Let $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ be real numbers. Show that

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n b_i^2 \right) \left(\sum_{i=1}^n a_i^2 \right) \quad (\text{Cauchy-Schwarz's inequality})$$

11. Let n be an integer, and let $x_1, x_2, \dots, x_n$ be real numbers such that

$$\sum_{i=1}^n x_i = \sum_{i=1}^n x_i^2 = n.$$

Show that

$$x_1 = x_2 = \dots = x_n = 1.$$

12. Let n be an integer, and let $x_1, x_2, \dots, x_n \in]0, 1[$ be real numbers such that

$$x_1 + x_2 + \dots + x_n < 1.$$

Show that

$$1 + \sum_{i=1}^n x_i \leq \prod_{i=1}^n (1 + x_i) \leq \frac{1}{1 - \sum_{i=1}^n x_i}. \quad (\text{Weierstras's inequality})$$

13. Let $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ be real numbers. Show that

$$\left(\sum_{i=1}^n (a_i + b_i)^2 \right)^{1/2} \leq \left(\sum_{i=1}^n a_i^2 \right)^{1/2} + \left(\sum_{i=1}^n b_i^2 \right)^{1/2}. \quad (\text{Minkowski's Inequality})$$

14. Let a, b be positive real numbers, and let $n \in \mathbb{N}$. Show that

$$(a+b)^n \geq a^n + na^{n-1}b.$$